

Contribution Journal

Cristian – Data Loading, Inspection, and Train/Test Split

Cristian worked on loading and inspecting the original stroke dataset using Python. This included checking the dataset size, columns, data types, missing values, duplicates, and the stroke target distribution. Cristian also separated the features and target and performed the required 70% training and 30% testing split. The training data was then prepared for EDA while keeping the testing data separate.

Khin Nyein Chan Thu – Exploratory Data Analysis and Visualization

Khin worked on exploratory data analysis using only the training dataset. This included examining numerical and categorical features and creating Matplotlib visualizations for variables such as age, BMI, glucose level, and stroke status. Khin also helped identify the large class imbalance between stroke and non-stroke cases and explained the patterns found during EDA.

Antonio – Data Cleaning, Preprocessing, and Feature Engineering

Antonio worked on cleaning and preprocessing the training data. This included handling missing BMI values, scaling and normalizing numerical features, encoding categorical variables, and using one-hot encoding. Antonio also worked on feature engineering by creating additional useful features from the existing data and helped prepare the dataset in numerical form for future machine learning models.

Steven – SMOTE, Final Validation, and Dataset Export

Steven worked on balancing the training dataset using SMOTE. This included comparing the class distribution before and after SMOTE and creating visualizations to show the improvement in class balance. Steven also checked the final dataset for missing values and other issues, reviewed the completed preprocessing results, and exported the final cleaned dataset for submission and future machine learning work.

Overall, all four members contributed to different parts of the project and worked together to make sure the dataset was properly cleaned, processed, balanced, and prepared for future machine learning models.